CONFÉRENCE

TRACK TECH

Web sémantique et langage naturel

Le cas du reporting européen

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Lead Engineer · e-Science Data Factory

01

Web sémantique

The Semantic Web : A new form of Web content that is meaningful to computers will unleash a revolution of new possibilities

By Tim Berners-Lee, James Hendler and Ora Lassila Scientific American ISSUE: May 2001



At the doctor's office, Lucy instructed her Semantic Web agent through her handheld Web browser. The agent promptly retrieved information about Mom's prescribed treatment from the doctor's agent, looked up several lists of providers , and checked for the ones in-plan for Mom's insurance within a 20-mile radius of her home and with a rating of excellent or very good on trusted rating services. It then began trying to find a match between available appointment times (supplied by the agents of individual providers through their Web sites) and Pete's and Lucy's busy schedules.

Web sémantique

Article Discussion

Le **Web sémantique**, le **Web 3.0** ou la **Toile sémantique**[1] est une extension du Web standardisée par le World Wide Web Consortium (W3C)[2]. Ces standards encouragent l'utilisation de formats de données et de protocoles d'échange normés sur le web, en s'appuyant sur le modèle *Resource Description Framework* (RDF).

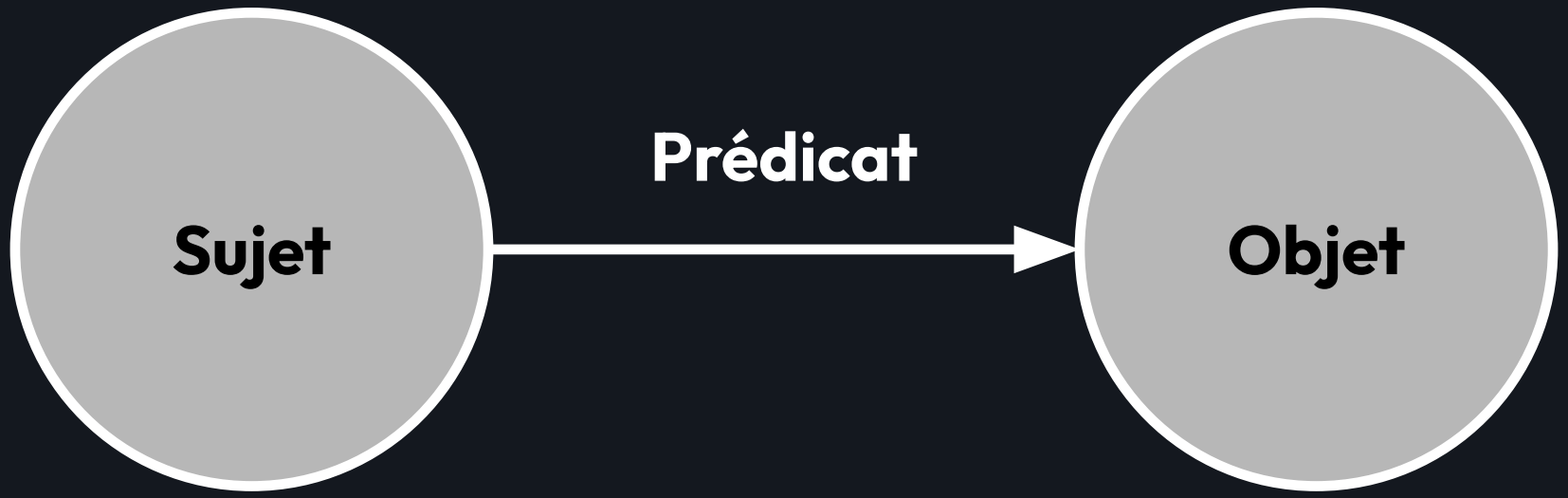


02

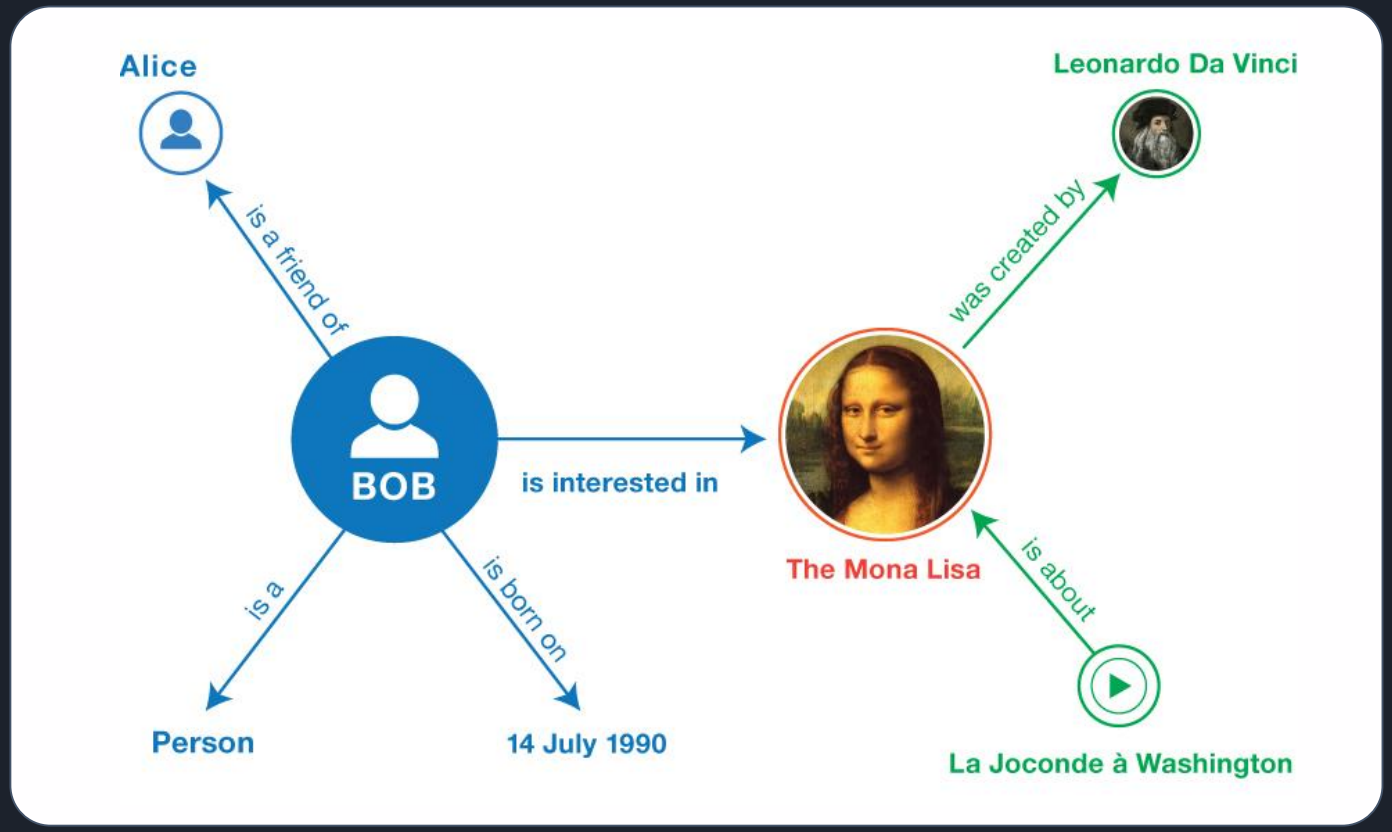
Graphe de données

Le triplet

Unité atomique d'information d’un graphe de données



Un exemple de graphe de données



03

Artefact sémantique

Definition

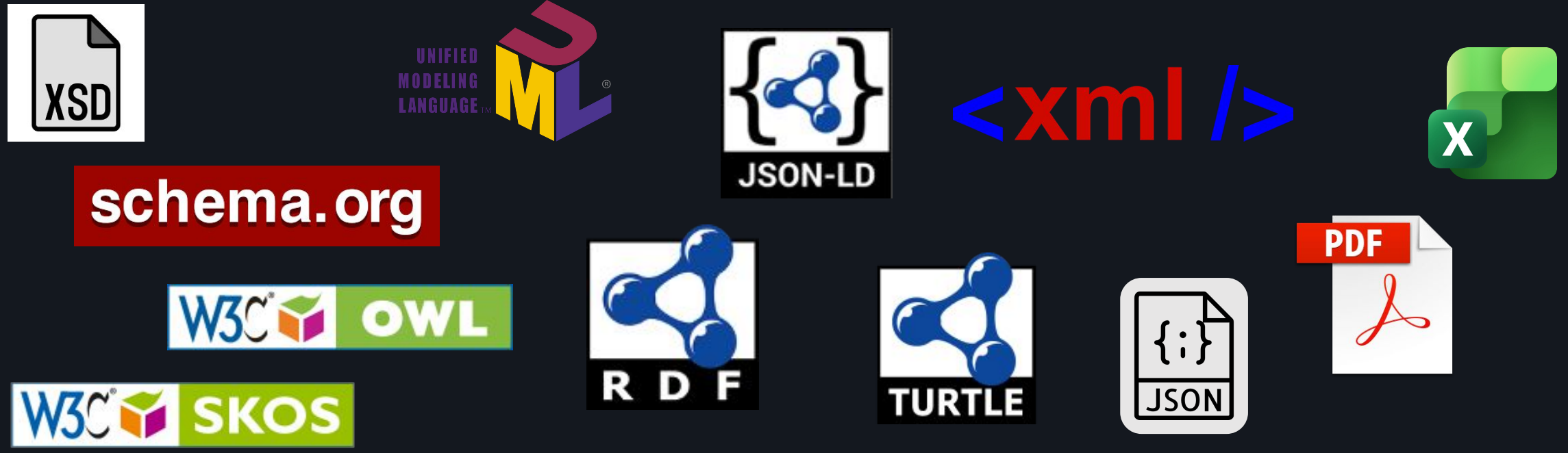
A **semantic artefact** is defined within our work as a **machine-actionable and -readable formalisation of a conceptualisation, enabling sharing and reuse by humans and machines**. These artefacts may have a broad range of formalisation, from loose sets of terms, **taxonomies**, **thesauri** to **higher-order logics**. Moreover, semantic artefacts are **serialised using a variety of digital representation formats.**

Yann Le Franc, Luiz Bonino, Hanna Koivula, Jessica Parland-von Essen, & Robert Pergl. (2022) : https://doi.org/10.5281/zenodo.6675295



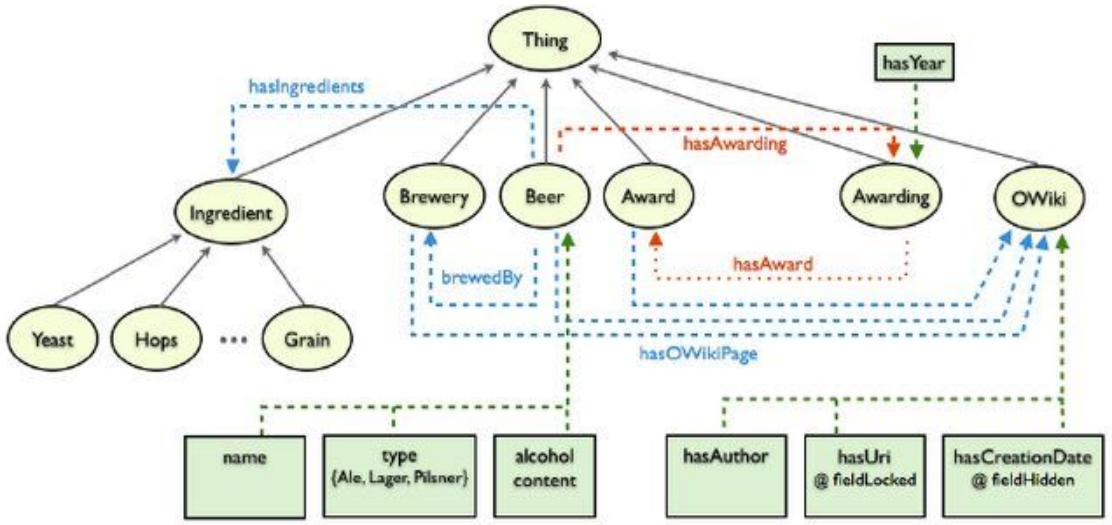
Quelles sont les représentations courantes ?

Une grande diversité de représentations numériques des modèles conceptuels

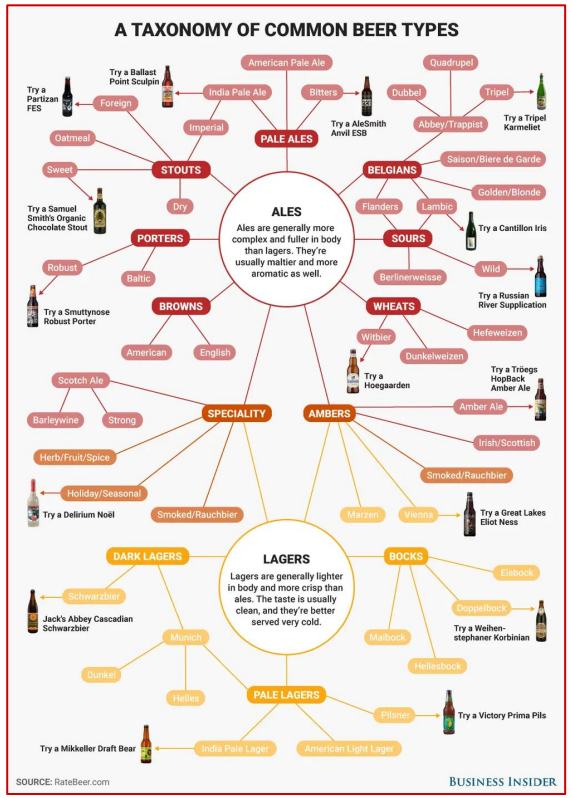


Que représentent les artefacts sémantiques ?

Modèles conceptuels du monde décrivant les concepts et leurs relations, nécessaires pour décrire les données et leur contexte.



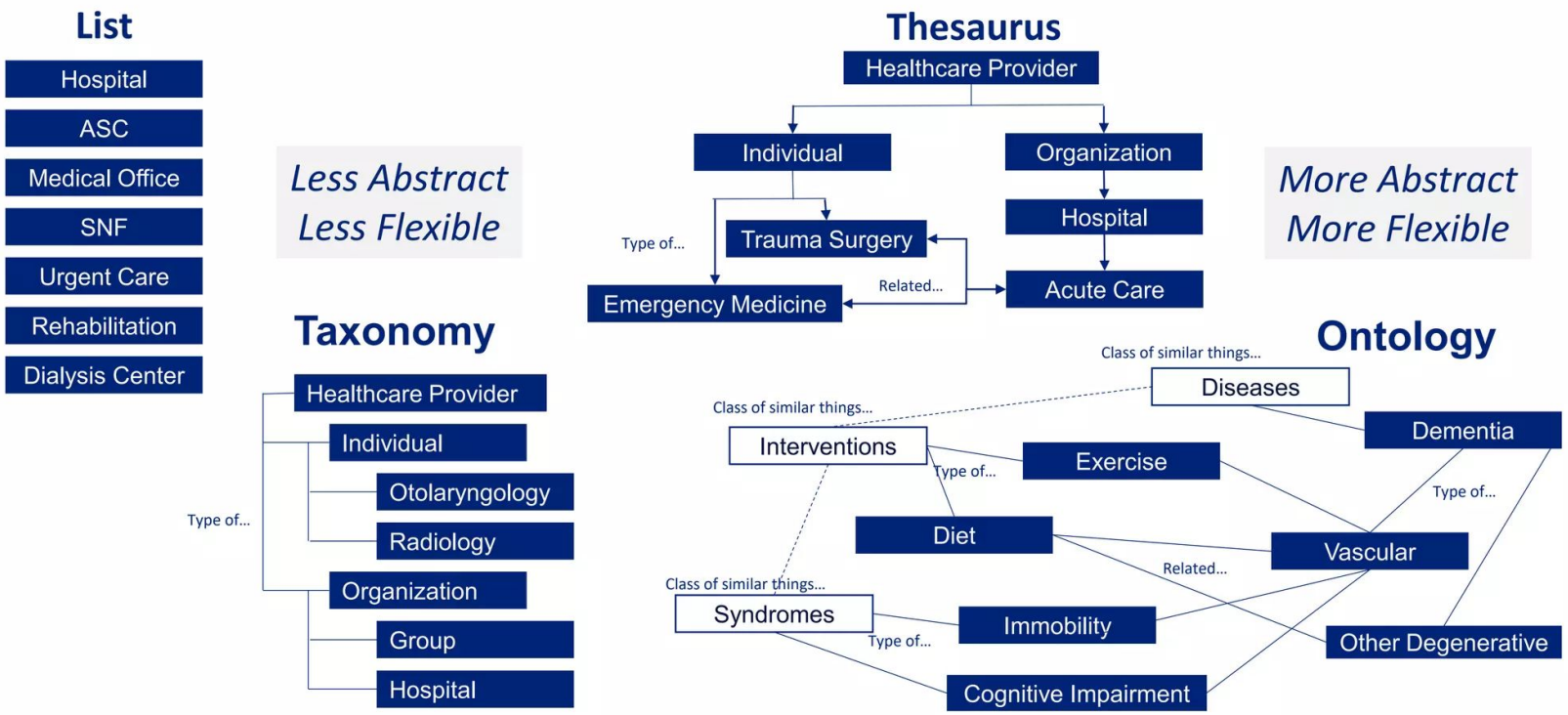
From Di Iorio, Angelo & Peroni, Silvio & Vitali, Fabio. (2012) - 10.1007/978-3-642-23172-8_24



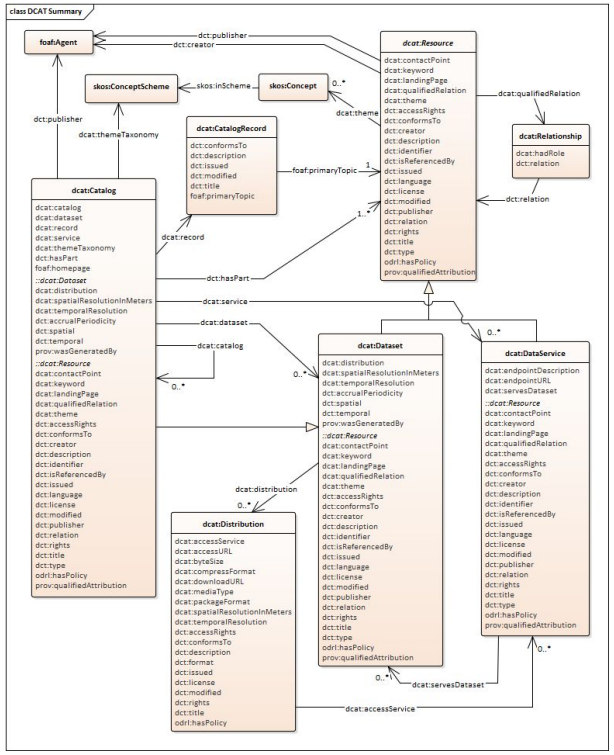


From https://www.businessinsider.com/different-types-of-beer-2016-13?r=US&IR=T

La diversité des artefacts sémantiques

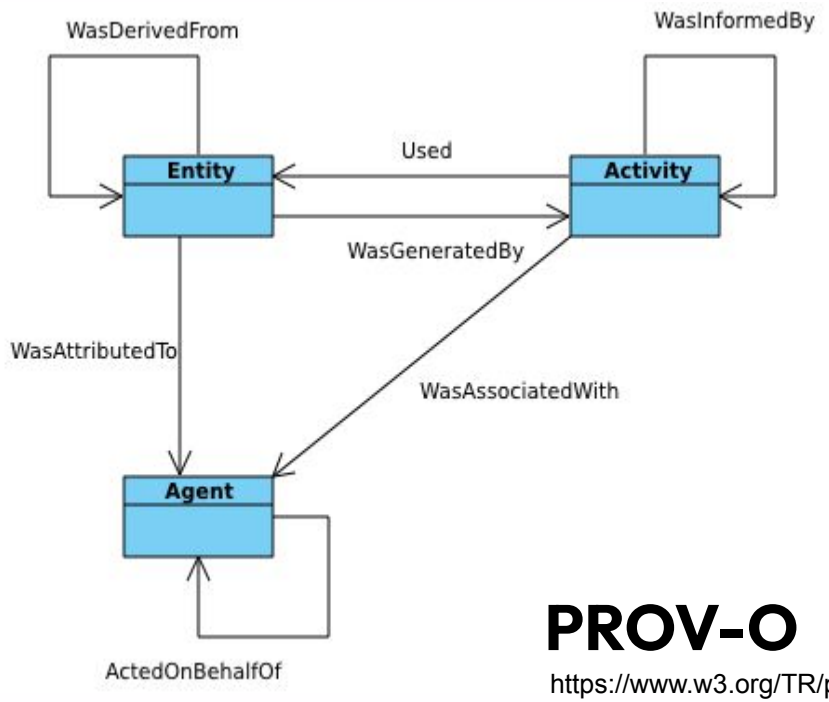


Exemples de modèles conceptuels génériques



DCAT

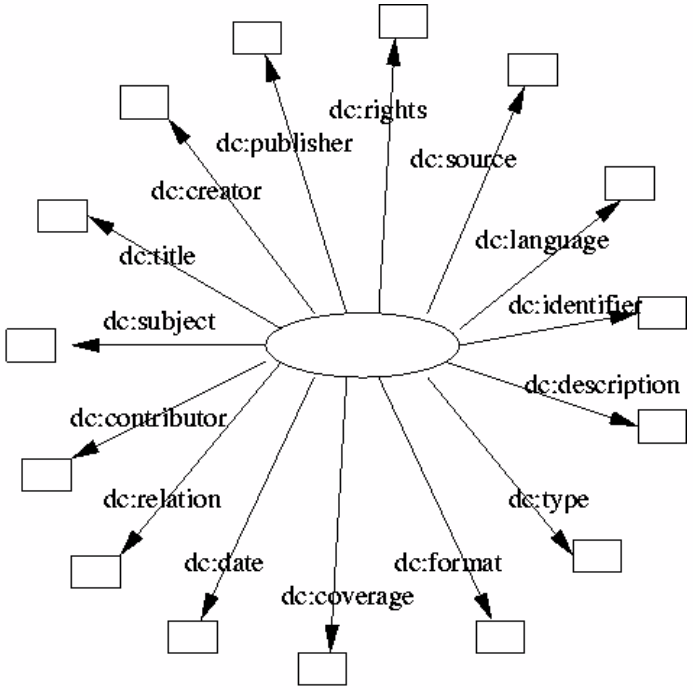
https://www.w3.org/TR/vocab-dcat-3/



PROV-O

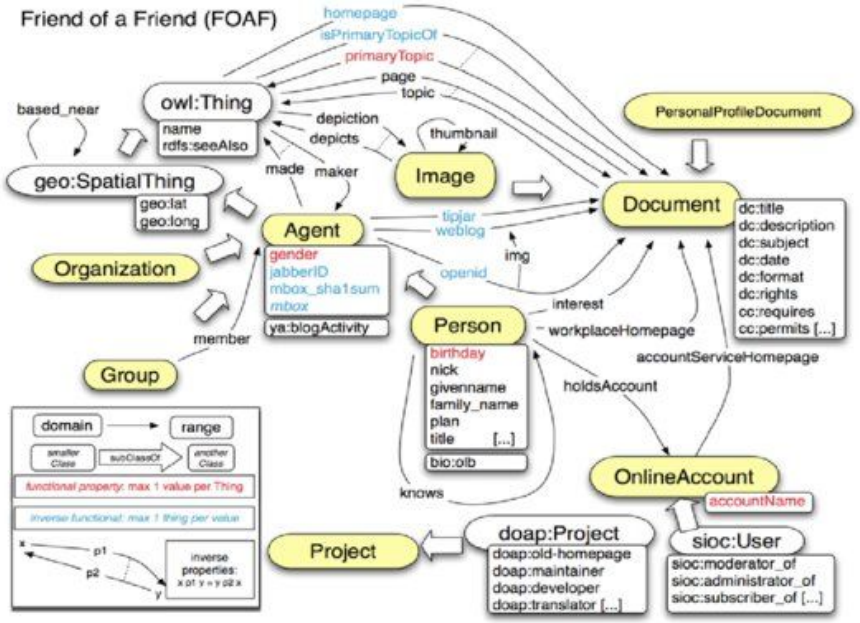
https://www.w3.org/TR/prov-o/

Exemples de modèles conceptuels génériques



DC

https://www.dublincore.org/specifications/dublin-core/



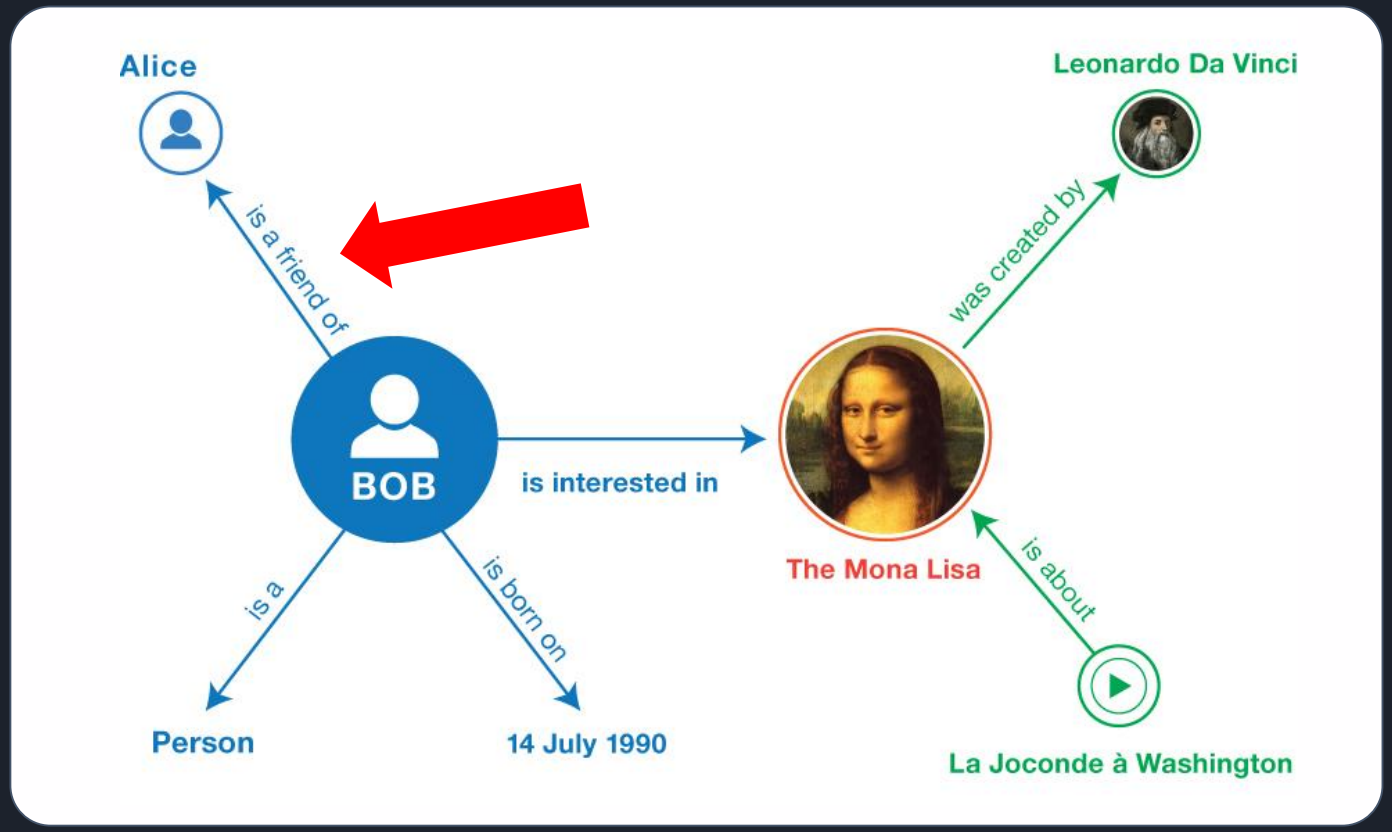
FOAF

https://xmlns.com/foaf/specs

03

Graphe de connaissance

Is a friend of



Is a friend of

RELATIONSHIP : Un vocabulaire pour décrire les relations entre les personnes

Friend Of

URI: http://purl.org/vocab/relationship/friendOf

A person who shares mutual friendship with this person.

Semantics

This property is symmetrical. Having this property implies being a foaf:Person. Every value of this property is a foaf:Person. It is a sub-property of Knows and owl:differentFrom.

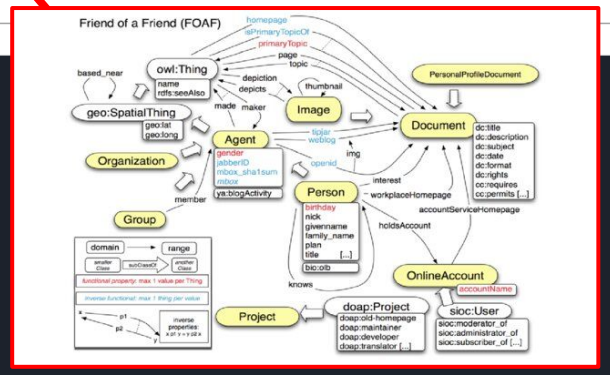
Status

- 2010-02-09 - semantic change by Ian Davis: Asserted that this property is a sub-property of owl:differentFrom, indicating that this is a relationship between different individuals

Other Information

Equivalent	msg3:friendOf

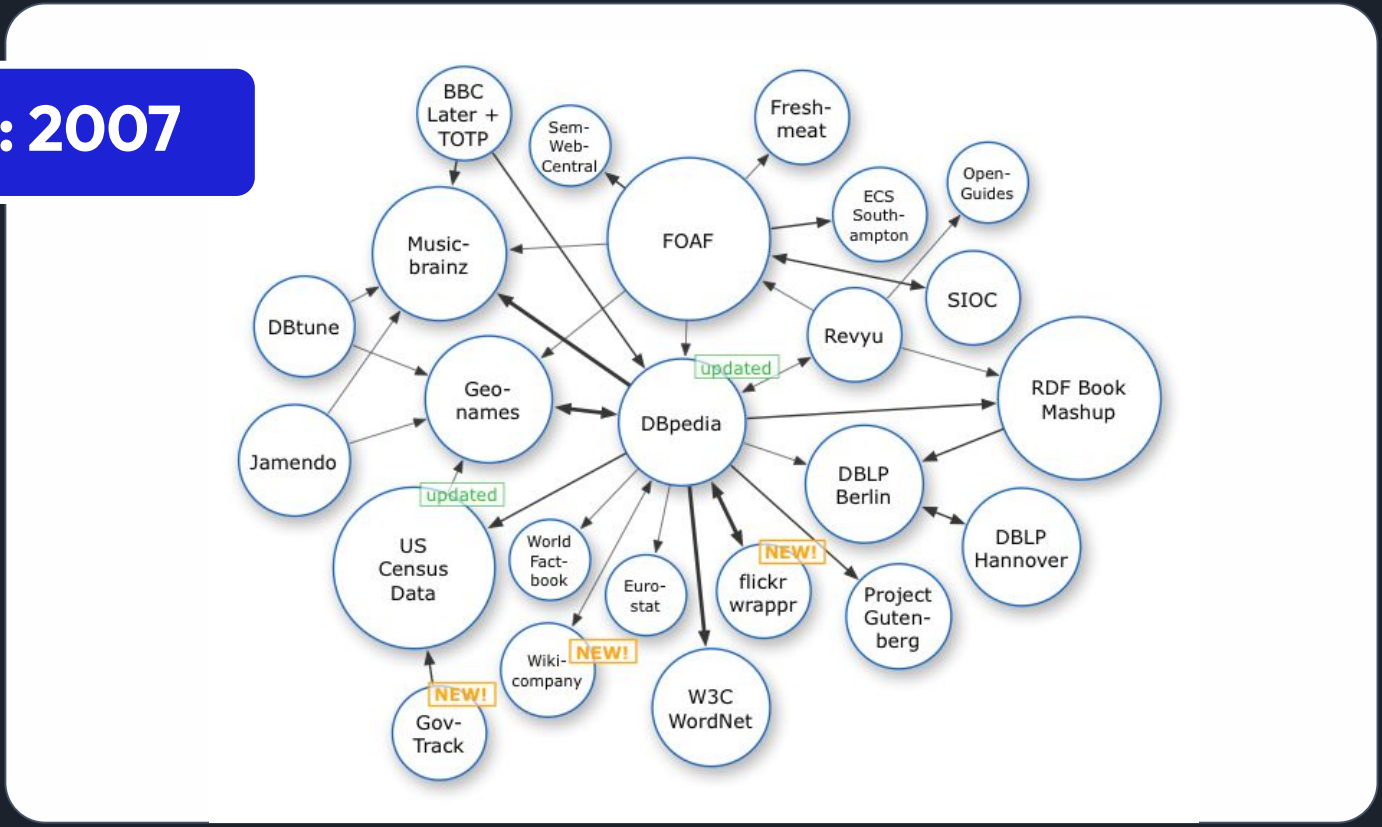
https://vocab.org/relationship



FOAF

Un exemple de graphe de connaissance

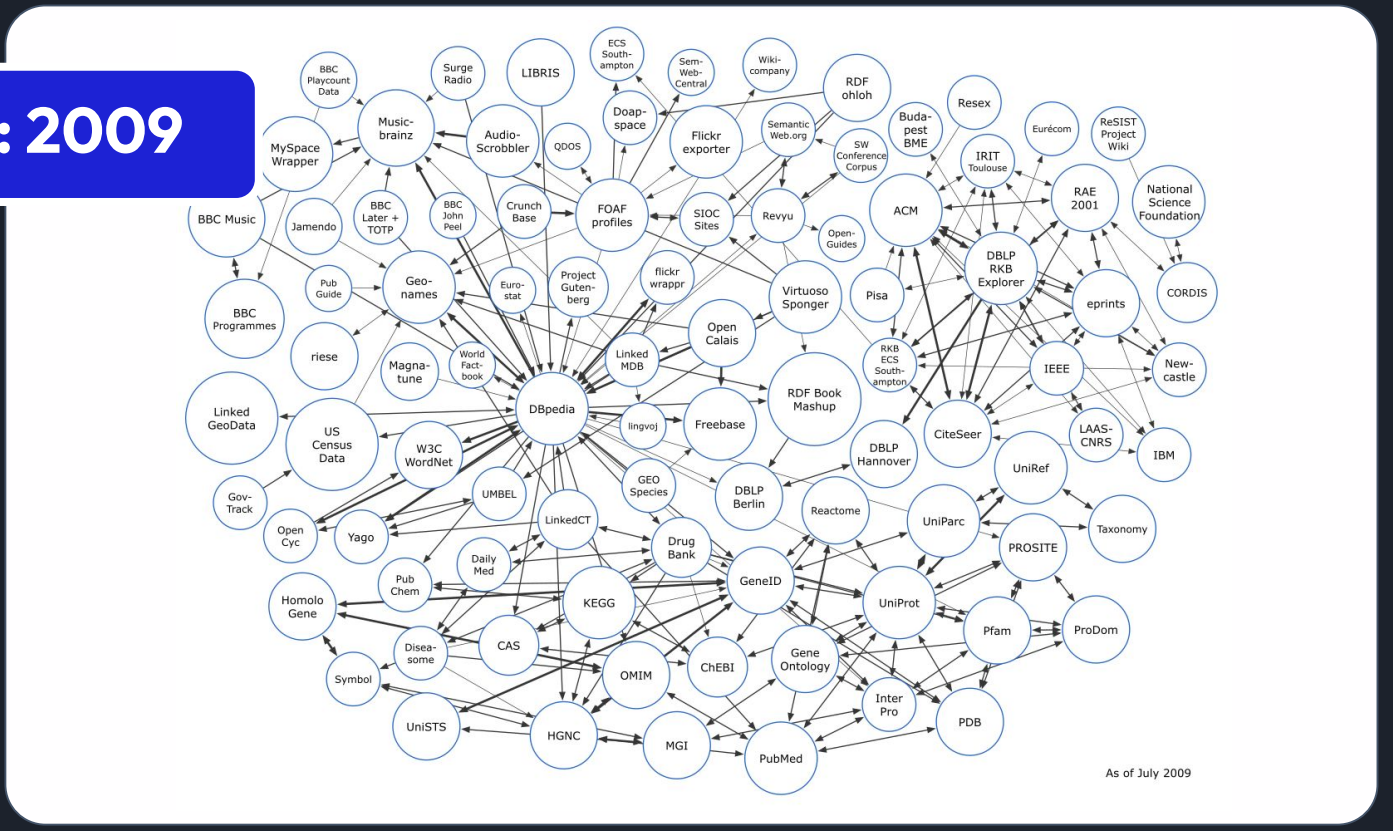
DBpedia : 2007



Représentation des relations entre les données ouvertes de **DBpedia** et divers autres projets du Web

Un exemple de graphe de connaissance

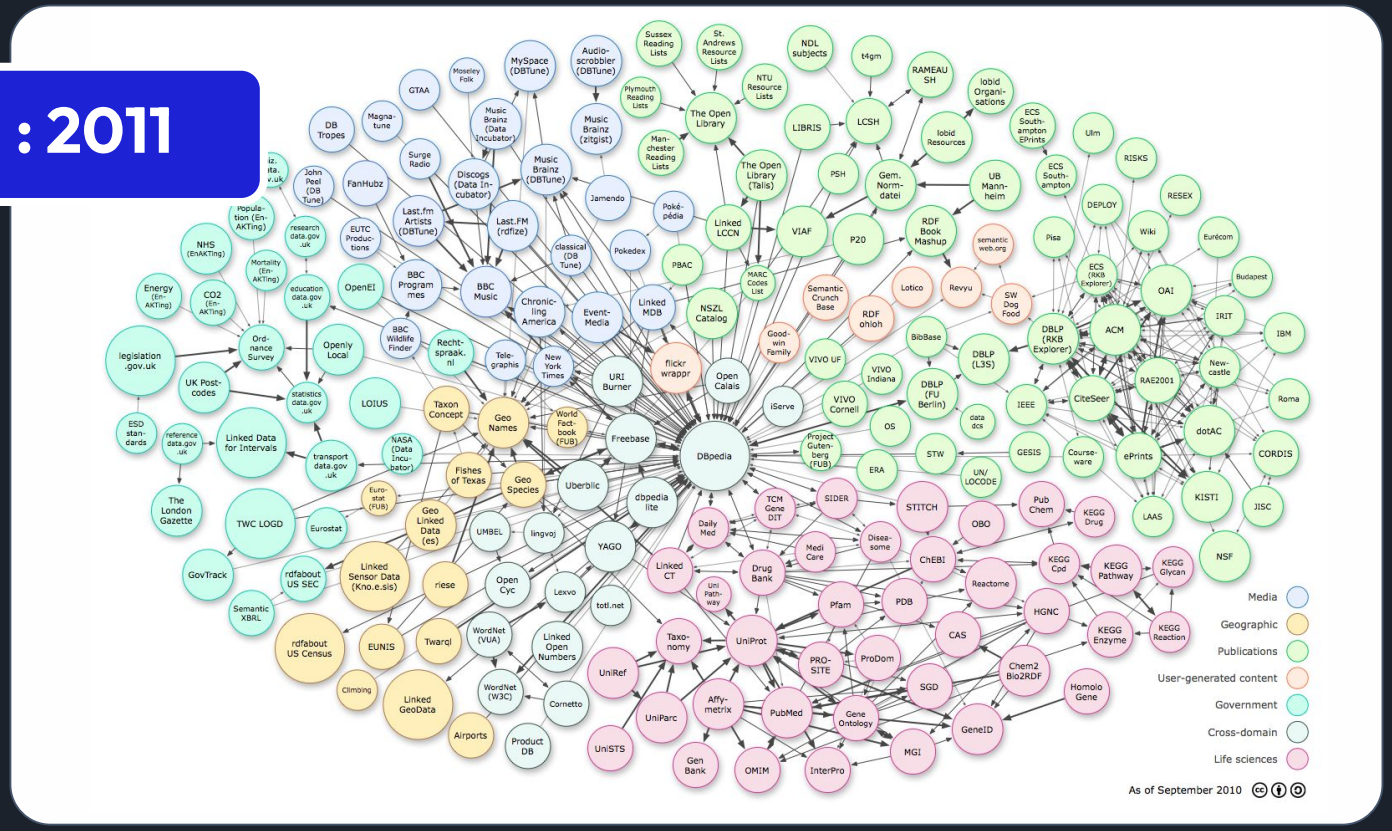
DBpedia : 2009



Représentation des relations entre les données ouvertes de **DBpedia** et divers autres projets du Web

Un exemple de graphe de connaissance

DBpedia : 2011



Représentation des relations entre les données ouvertes de **DBpedia** et divers autres projets du Web

En juin **2021**, **DBpedia** contenait plus de **850 millions de triplets**.

04

Reporting Européen

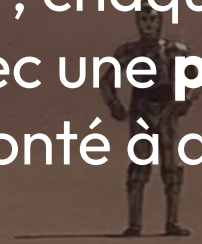
Combien de temps Manuel a-t-il passé au développement de la plateforme LUMIS pour la période de reporting ERP2 ?



C-3PO [projet interne]

Objectif

La **finance** (Margaux) et la **gestion de projet** (Florian) peut **générer les** chiffres des **rapports périodiques** sur les projets européens via des **requêtes en langage naturel** dans l'outil C-3PO ; chaque **donnée** est **traçable** jusqu'aux transactions sources et **validée** avec une **précision de 100 %** sur la base d'un jeu de requêtes de référence confronté à des enregistrements vérifiés.



https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

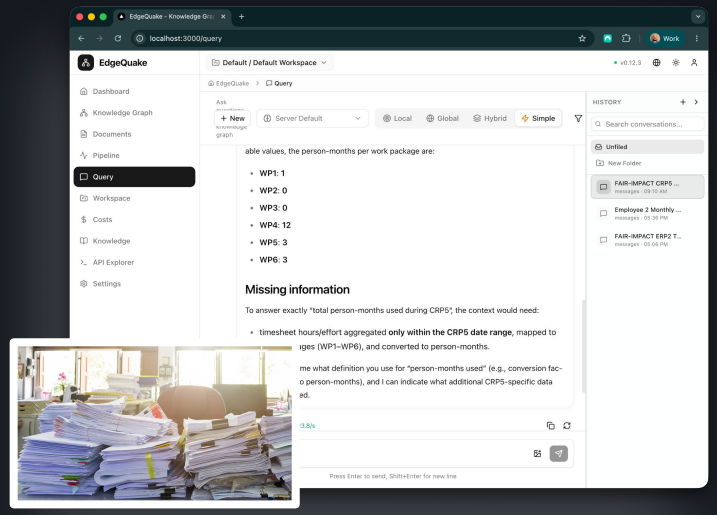
2025/26 - Prototypage



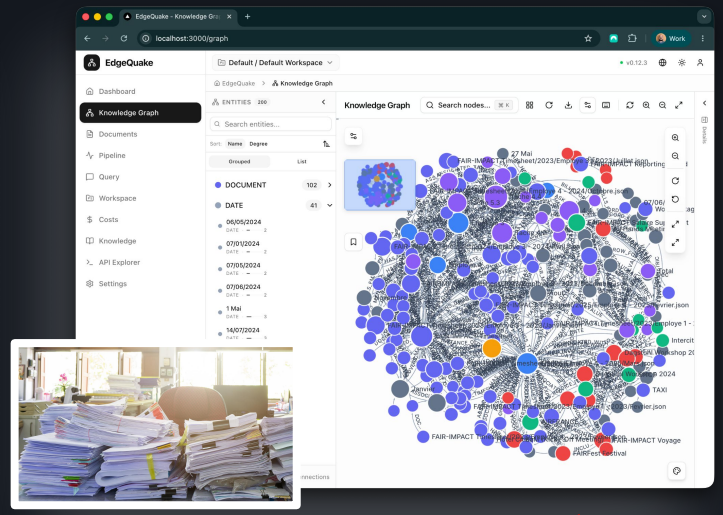
https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

2025/26 - Prototypage

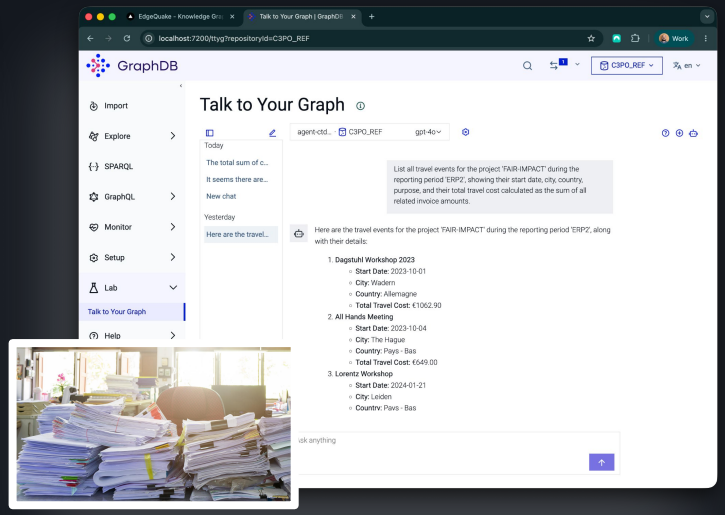
RAG



GraphRAG



LLM + **Graphe de connaissance**



~~Graphe de connaissance~~

Ingestion via ETL

précision

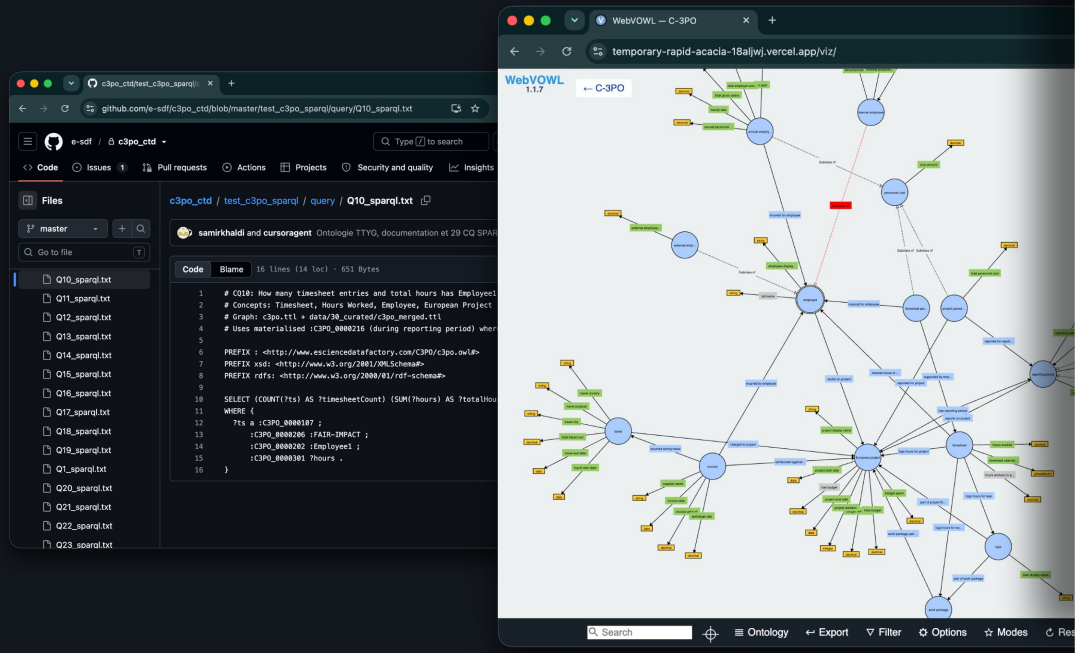
0 %

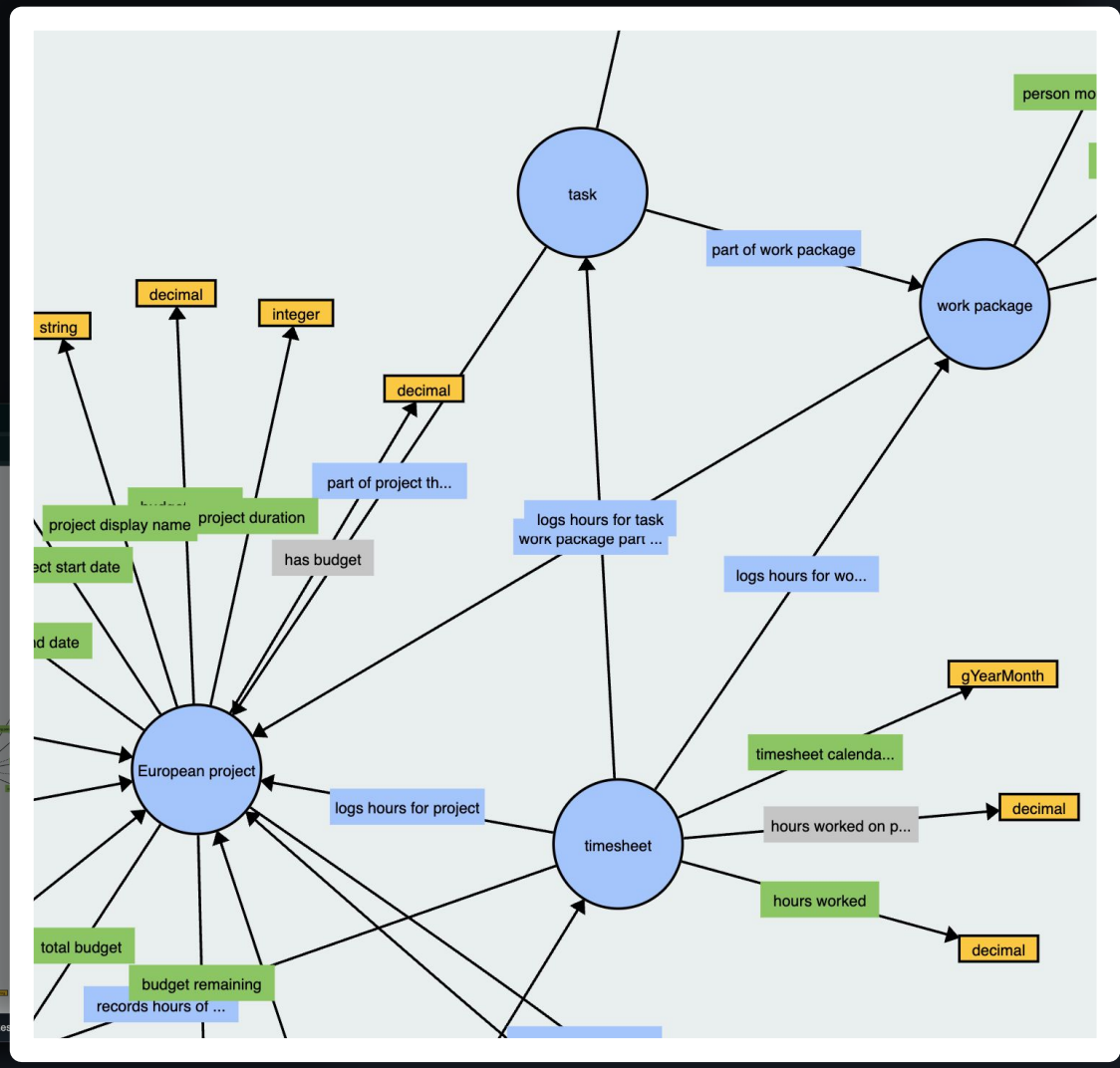
100 %

Ontologie C-3PO

Version 1.0

Modèle conceptuel décrivant les données et leurs relations dans le contexte du reporting des projets Européens.





Construire une ontologie

Lot Methodology

https://lot.linkeddata.es/

SAMOD

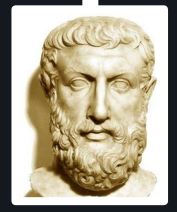
An agile methodology for the development of ontologies

https://essepuntato.it/samod/

Competency Questions

Domain Expert





Ontologist

Exemple

Video Game Ontology - Competency Questions

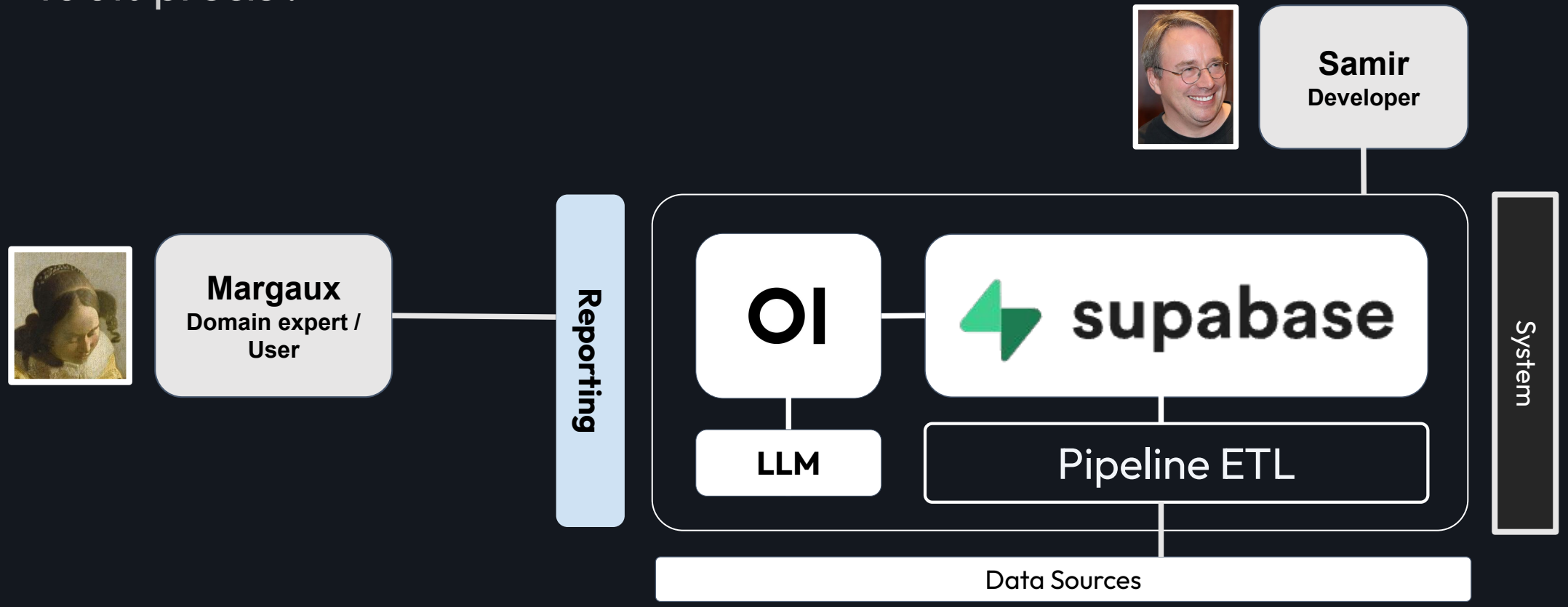
Game Metadata

1. What is the username of the player? Japskua, TheGuy47
1. Who are the friends of the player? “Ivanhoe444, Super1337”
2. Who are the friends that play other games as well with this player? “Japskua, C00LDud3”
3. Who are the most active players in the game? “Michael98, Justin Bieber”
4. What are the achievements of my friends? “Cool Guy”, “Mr. Guru”
5. Who does the player play with? “Nobody”, “ArnoldTheHunkz”
6. What are the achievements your friend has received? “Guru” “Kill’Em’All”
7. What achievements does a game have? “Unlock me!”, “You Rock”
8. How many hours has this game been played in total? 50 000 hours
9. What are the types of achievements in this game? Skill-based, Veteran, Luck,
10. What are the types of achievements a game can have? Skill-based, Veteran, Luck,
11. What is the genre of the game? RPG, Adventure, Puzzle
12. What items exist in a game: “Magical Staff”, “Ferrari”
13. Who is the creator of the game? “Electronic Arts”, “Janne Parkkila”
14. What is the release date of the game? 14.05.2005
15. What are the games similar to this one? “Hitman 1, Hitman 2, MassEffect 2”
16. List all games of a certain Genre? “NHL12, Fifa13, NBA10”


2026/27 - Industrialisation
en cours, à vos risques et périls
https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

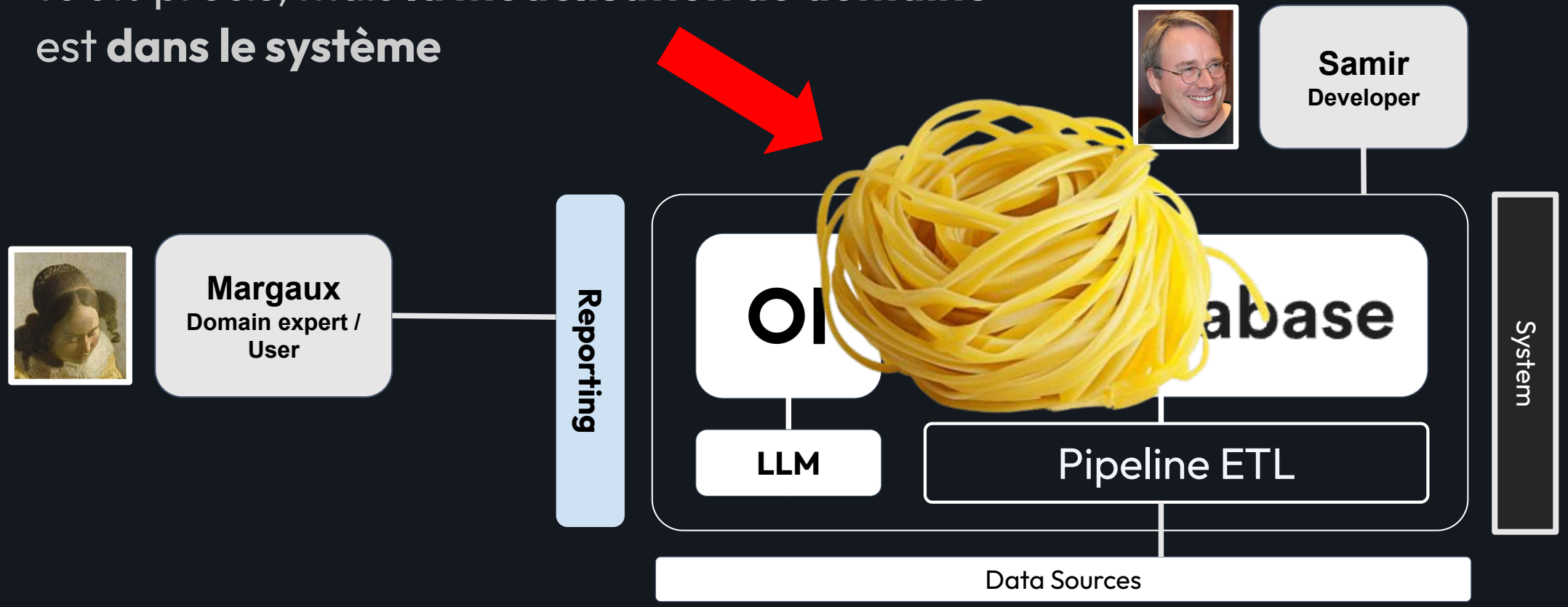
Une architecture pragmatique

100% précis !



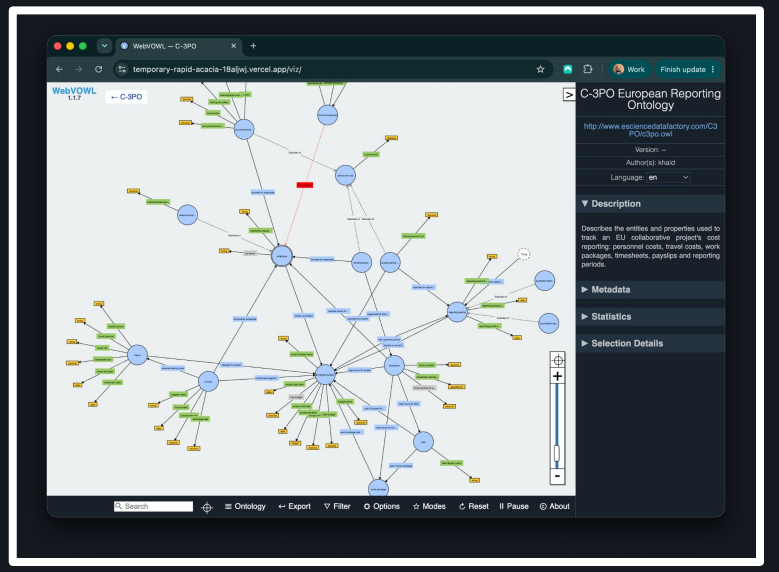
Une architecture pragmatique

100% précis, mais **la modélisation du domaine** est **dans le système**



La modélisation du domaine : élément architectural





https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

La modélisation du domaine : élément architectural

C-3PO must be designed as **Ontology-driven system** and follow those 4 rules :

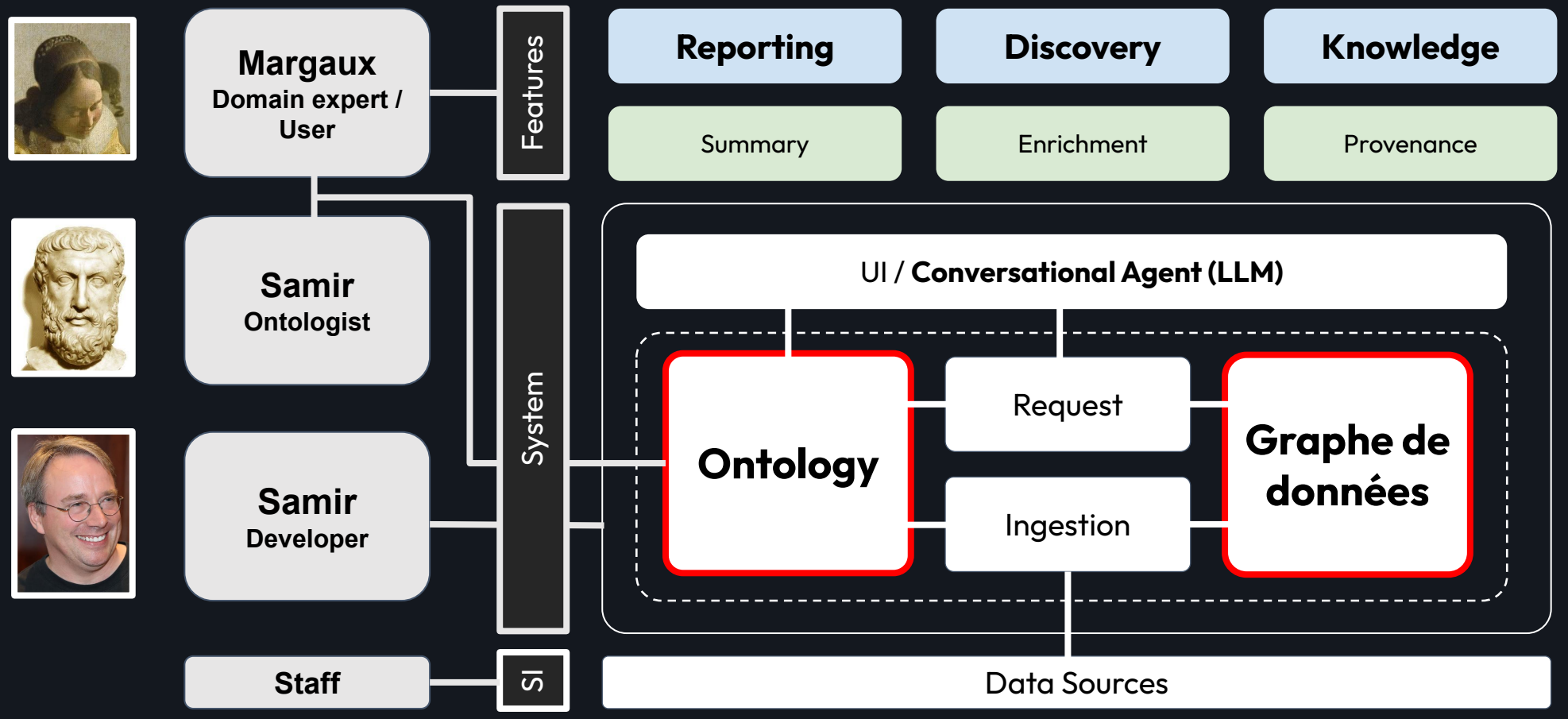
1. The ontology defines the domain: structure and business logic, distinct from each other, none of it in code.
2. Anything not expressible as structure or rule is an exception, and must have its own explicit contract.
3. Consequence of 2: the system must verify it still works correctly after any ontology change against that contract.
4. Every rule must be deterministic: same input, same output, always.

Nice to have (implement if low-cost)

- Support two rule-set versions running concurrently during transition periods

https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

Architecture fonctionnelle cible



Work in progress...

https://www.dublincore.org/specifications/dublin-core/dcq-rdf-xml/2001-08-29/

Merci

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